

Practice Test III for in Class (on classes 18-24).

Convert the following decimals to fraction form, then to percent form.

① a) 0.6

b) 0.587

c) 0.2000

d) 38.0002040

Convert the following fractions to decimal form, then to percent form.

② a) $\frac{3}{10}$

b) $\frac{7}{10000}$

c) $\frac{843}{10000000}$

d) $\frac{304016}{100,000,000}$

③ a) $\frac{1}{2}$

b) $\frac{3}{5}$

c) $\frac{1}{4}$

d) $6\frac{3}{4}$

④ $\frac{5}{8}$

⑤ $\frac{7}{9}$

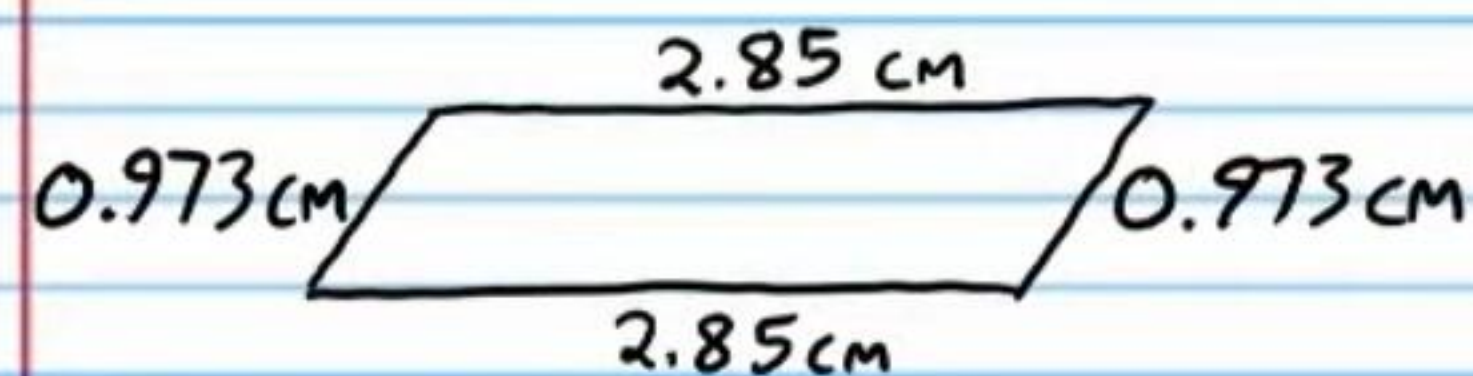
⑥ $\frac{8}{27}$

⑦ $826\frac{1}{11}$

⑧ $\frac{3}{7}$

⑨ A cat weighs 7 pounds. Its weight increases by 22%. What is its new weight?

⑩ Find the perimeter of the parallelogram below.



⑪ The peregrine Falcon is the fastest animal on the planet. It dives at 240 mph. If this bird dives for 0.00573 hours, how far would it travel during this time?

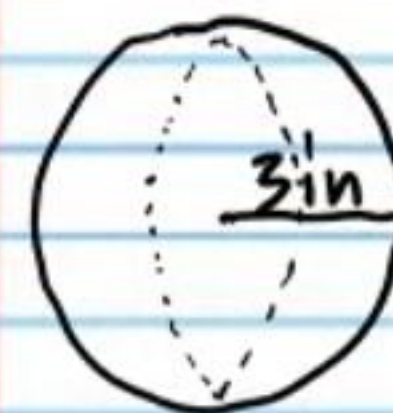
⑫ Billy Bob had \$18.07 but he spent \$9.23 on popcorn. How much does he have left?

⑬ Convert the percents below to decimals, then to fractions.

a) 23%	b) 0.8701%
c) 7.8%	d) 522.43%

⑭ You deposit \$90 in a savings account that pays 0.8% simple interest annually. If you leave the money in the account for three years, how much will you make in interest and what will be your total account sum?

⑮ The approximate volume of a sphere is found using the following formula: $V \approx 4 \times 3.14 \times r^3 \div 3$. Find the volume of the sphere below.



⑯ $\frac{1}{5}$ of a substance is iron. What percent of the substance is iron? What fraction of the substance is not iron? Write this fraction as a percent.

⑰ 8.4% of a seven pound sample is lithium. How many pounds of this sample are lithium?

18) 0.04 percent of the human body is sulfur. If 0.076 pounds of a man are sulfur, how much does the man weigh?

19) If there are 1.8 grams of sugar in an eight gram piece of candy, what percent of the piece of candy is sugar?

20) Perform the following operations.

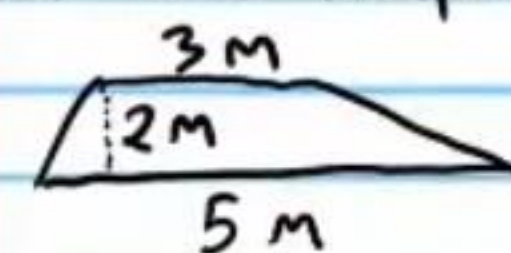
a) $2 \times (5-2)^2 + 49 \div 7$

b) $6 \times [5^2 - 20 \div (10-6)] + 1$

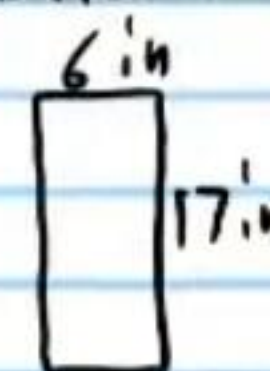
c) $16 - 9 - 5 + 2 + 1$

d) $16 \div 2 \times 2 \div 4$

21) a) The area of a trapezoid is found with the following formula: $A = (b_1 + b_2) \times h \div 2$, where b_1 and b_2 are the bases and h is the height. Find the area of the trapezoid below.



b) The perimeter of a rectangle is found with the following formula: $P = 2 \times W + 2 \times L$. Use this formula to find the perimeter of the rectangle below.



22) You borrow \$312 from your sister and she charges you 8% simple interest daily. How much money in interest will you owe her after four days? What is the total amount of money you will owe her after four days?

(23) A store gives a 20% discount on all items. If a customer buys a hat listed for \$ 8.25, how much will he pay before tax? If the customer has to pay a 10% sales tax, how much will he pay after tax?

(24) If a number is decreased by 43% and the result is 2.166, what was the original number?

(25) Gordon had 24 million dollars. His investors gave him an additional 8 million. His funds increased by what percent?